Project Documentation

Clinic Scheduling System – Group 20c “The winners”

System name	Clinic Scheduling System
Group number	20c
Team members (mark the group manager)	Gazi Ataol (s34024) – group manager; Yryskeldi Omuraliev (s29039); Nikita Antonovitch (s31402); Nazar Martyniuk (s30822)
Client team (group no.)	
Project repository (from week 6)	
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Revision history

Version	Date	Author(s)	Changes
1.0	15.10.2026	G. Ataol, Y. Omuraliev, N. Antonovitch, N. Martyniuk	First version (Assignment 2) – Part I, sections 1–6

Document structure

Section	Content	Added in
1–6	Part I – Software Requirements Specification (incl. user stories)	Assignment 2 (week 2)
7	Use case model	Assignment 3 (week 3)
8	Class diagram	Assignment 4 (week 4)
9	Use case point estimation	Assignment 5 (week 5)
10	Dynamic analysis	Assignment 7 (week 7)
11	User interface prototype	Assignment 8 (week 8)
12	Test plan	Assignment 10 (week 10)
13	Acceptance tests	Assignment 12 (week 12)
14	Implementation	Assignment 13 (week 13)

Part I – Software Requirements Specification

1. Introduction

1.1 Purpose

What is this document for and who will read it (e.g. the client team, the development team, testers)?

1.2 Scope

Name the system and describe its main goal in a few sentences. State what the system will do and, just as importantly, what it will not do (e.g. "The system supports online reservation of books. Purchasing new books is not part of the system.").

1.3 References

The documents this specification is based on: the client brief, the minutes of the initial interview, notes from the client meeting.

1.4 Glossary

Minimum 6 domain terms. Define words the client uses that an outsider might not understand, and every abbreviation used in this document.

Term	Definition
Reservation	*A reader's request to borrow a specific book as soon as a copy becomes available.*

2. Overall description

2.1 Product description

The client is a medical clinic with one location and several doctors of different specialties. Today receptionists handle appointments by phone and email, and the schedule is kept on paper and in separate calendars. Checking which doctor is free takes a lot of time, double bookings happen, patients forget their visits because nobody reminds them, and patients cannot see free times without calling the clinic.

The Clinic Scheduling System keeps one central schedule for the whole clinic. The clinic administrator registers doctors with their specialties, defines the medical services and sets the working hours of each doctor. Doctors can mark the periods when they are not available. Patients create an account, search for a doctor by specialty, see the free time slots and book, reschedule or cancel their own appointments. Receptionists do the same for patients who call or come to the clinic, and they can search for patients, doctors and appointments. Doctors see their daily and weekly schedule and, before each visit, the contact details of the patient and the basic information needed for the visit. The system does not allow two patients to be booked with the same doctor at the same time. It records the status of every appointment and sends reminders and notifications about upcoming, cancelled and rescheduled appointments.

Receptionists, doctors and the administrator use the system in the clinic on desktop computers. Patients use it from home on computers and mobile phones through a web browser. Notifications are sent through an external email service. Sending SMS messages is still an open issue (see section 6). No connection with other clinic systems is planned for this version.

Online payments, video consultations and electronic prescriptions were named by the client as possible future extensions. They are not part of this version.

2.2 User classes

The system has four user classes. The same names are used in the user stories (section 5) and in the use case model (section 7).

User class	Description	Main needs
Patient	A person registered in the clinic who has an account in the system.	Find a doctor by specialty, see free time slots, book, reschedule and cancel own appointments, see own past and upcoming appointments, receive reminders.

User class	Description	Main needs
Receptionist	Clinic employee at the front desk who handles patients by phone and in person.	Register patients, book, reschedule and cancel appointments on behalf of patients, check doctors' availability, search patients, doctors and appointments, mark missed appointments.
Doctor	Medical specialist employed by the clinic.	See own daily and weekly schedule, see patient information before a visit, mark own unavailability, mark appointments as completed.
Clinic Administrator	Clinic manager responsible for the organisation of the clinic and for the system.	Register doctors and their specialties, define medical services, manage working schedules, manage user accounts and roles, view reports about cancelled and missed appointments.

2.3 Constraints and assumptions

ID	Type	Description
C-01	Constraint	The system must process patients' personal data in accordance with the GDPR (EU 2016/679).
C-02	Constraint	Patient information may be accessed only by authorised users according to their role (clinic rule from the client brief).
C-03	Constraint	The system must be usable from a web browser on both desktop computers and mobile devices.
A-01	Assumption	Every patient has an e-mail address that can be used for notifications and reminders.
A-02	Assumption	The system manages exactly one clinic location; departments and additional locations may be added in the future.
A-03	Assumption	Exactly one doctor is assigned to an appointment.
A-04	Assumption	The standard duration of a time slot is 30 minutes; services that need more time use a multiple of this duration.

3. Functional requirements

The functional requirements below describe what the system shall do. The source column refers to the documents listed in section 1.3: the client brief [1], the interview questions [2] (Q1–Q10) and the client meeting [3].

ID	Requirement	Priority	Source
FR-01	The system shall allow a receptionist to register a patient with name, date of birth, phone number and email address.	Must	Client brief, §4, §5
FR-02	The system shall allow a patient to create an account with name, date of birth, phone number and email address.	Must	Client brief, §5
FR-03	The system shall allow the clinic administrator to register a doctor with name, specialty, contact details and weekly working hours.	Must	Interview, Q2
FR-04	The system shall allow the clinic administrator to define a medical service with a name and a duration in minutes.	Must	Interview, Q4
FR-05	The system shall allow a doctor to mark a period as unavailable.	Should	Client brief, §4
FR-06	The system shall allow a patient to search for doctors by specialty.	Must	Interview, Q10
FR-07	The system shall display the free time slots of a selected doctor for a selected service.	Must	Interview, Q10
FR-08	The system shall allow a patient to book an appointment in a free time slot.	Must	Client brief, §4
FR-09	The system shall allow a receptionist to book an appointment on behalf of a patient.	Must	Client brief, §4
FR-10	The system shall refuse to book two appointments with the same doctor that overlap in time.	Must	Client brief, §4
FR-11	The system shall allow a patient or a receptionist to reschedule a scheduled appointment to another free time slot.	Must	Client brief, §4
FR-12	The system shall allow a patient or a receptionist to cancel a scheduled appointment.	Must	Client brief, §4
FR-13	The system shall record the status of every appointment as Scheduled, Completed, Cancelled or Missed.	Must	Client brief, §5; Interview, Q3
FR-14	The system shall display to a patient the list of their own past and upcoming appointments with the status of each.	Must	Interview, Q3

ID	Requirement	Priority	Source
FR-15	The system shall display to a doctor their appointments in a daily view and in a weekly view.	Must	Client brief, §4
FR-16	The system shall display to a doctor, for each appointment, the patient's name, contact details, appointment details and basic medical information for the visit.	Should	Interview, Q7
FR-17	The system shall send a reminder to the patient 24 hours before a scheduled appointment.	Should	Interview, Q5; Client brief, §6
FR-18	The system shall send a notification to the patient when their appointment is cancelled.	Should	Interview, Q5
FR-19	The system shall send a notification to the patient when their appointment is rescheduled.	Should	Interview, Q5
FR-20	The system shall allow the clinic administrator to create, modify and deactivate user accounts and assign a role to each account.	Must	Client brief, §4

Priority (MoSCoW): Must – the system is useless without it; **Should** – important, but a workaround exists; **Could** – nice to have; **Won't** – agreed not to be delivered this time.

4. Non-functional requirements

Each non-functional requirement states how it will be measured or checked.

ID	Category	Requirement	How it is verified
NFR-01	Performance	The list of free time slots of a doctor shall be displayed within 2 seconds for 95% of requests with up to 50 users logged in at the same time.	Load test with 50 simulated users; response times measured
NFR-02	Security	Every function of the system shall be accessible only to the user classes listed for it in section 3; a request from an unauthorised role shall be rejected with an error message.	Test cases: each function tried from every role
NFR-03	Security	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Test case: 5 wrong passwords, then a correct one
NFR-04	Security	Passwords shall be stored only in hashed form; no plain-text password shall exist in the stored data.	Inspection of the stored data
NFR-05	Availability	The system shall be available at least 99% of the time during clinic opening hours (Mon–Fri 08:00–20:00).	Monitoring report for one month

ID	Category	Requirement	How it is verified
NFR-06	Compatibility	All functions shall work in the current versions of Chrome, Firefox, Edge and Safari on screens from 360 px (mobile) to 1920 px (desktop) wide.	Manual test of every function in each browser and both screen sizes
NFR-07	Usability	A patient who has never used the system shall be able to book an appointment within 5 minutes without help in at least 8 of 10 usability test sessions.	Usability test with 10 participants
NFR-08	Reliability	At least 99% of reminders shall be sent within 10 minutes of their scheduled sending time.	Comparison of the notification log with the scheduled times over one month
NFR-09	Accessibility	All pages used by patients shall satisfy WCAG 2.1 level AA.	Automated accessibility check (e.g. axe) plus manual keyboard-only test

5. User stories

One user story per team member. Each story uses a user class from section 2.2, has at least three acceptance criteria (including an error or edge case) and is linked to the functional requirements of section 3.

US-01 – Book an appointment with a doctor of a chosen specialty

ID	US-01	Author	Gazi Ataol
Title	Book an appointment with a doctor of a chosen specialty	Priority	Must
Related requirements	FR-06, FR-07, FR-08, FR-10	Estimate (optional)	5 story points
Story	As a patient, I want to search for a doctor by specialty and book one of their free time slots, so that I can get an appointment without calling the clinic.		

Acceptance criteria	AC1 Given I am logged in as a patient, when I select the specialty “Dermatology”, then the system shows the list of doctors with that specialty. AC2 Given I have selected a doctor and a service, when I open the calendar, then only the doctor’s free time slots for the next 30 days are shown. AC3 Given I have selected a free time slot, when I confirm the booking, then the appointment is saved with status Scheduled and appears in my list of upcoming appointments. AC4 (edge case) Given another patient booked the same slot a moment before I confirmed, when I confirm, then the system refuses the booking, shows the message that the slot is no longer free and refreshes the list of free slots.
Notes / open questions	How far in advance can a patient book (30 days? 90 days?).

US-02 – Reschedule an appointment for a patient

ID	US-02	**Author**	Yryskeldi Omuraliev
Title	Reschedule an appointment for a patient	**Priority**	Must
Related requirements	FR-09, FR-10, FR-11, FR-19	**Estimate (optional)**	3 story points
Story	As a receptionist, I want to move a patient's scheduled appointment to another free time slot, so that the patient keeps their visit when the original time no longer suits them or the doctor.		
Acceptance criteria	AC1 Given I have opened a scheduled appointment, when I choose "Reschedule", then the system shows the free time slots of the same doctor for the same service. AC2 Given I have selected a new free slot, when I confirm, then the appointment keeps its ID, gets the new date and time, stays in status Scheduled and the patient receives a rescheduling notification. AC3 (error case) Given the selected new slot overlaps with another appointment of the same doctor, when I confirm, then the system refuses the change and the appointment keeps its original time. AC4 (edge case) Given the appointment has status Completed, Cancelled or Missed, when I open it, then the "Reschedule" action is not available.		
Notes / open questions	Should the patient be asked to confirm a reschedule made by the receptionist?		

US-03 – Mark a period as unavailable

ID	US-03	**Author**	Nikita Antonovitch
Title	Mark a period as unavailable	**Priority**	Should
Related requirements	FR-05, FR-07, FR-12, FR-18	**Estimate (optional)**	3 story points
Story	As a doctor, I want to mark days or hours in which I am unavailable, so that patients cannot book me when I am on leave or at a conference.		
Acceptance criteria	AC1 Given I am logged in as a doctor, when I mark 20–24 October as unavailable, then no time slots of mine are shown as free on those days to patients and receptionists. AC2 Given I mark a period in which I have no appointments, when I save it, then the period is stored and shown in my weekly view. AC3 (edge case) Given I mark a period in which I already have scheduled appointments, when I save it, then the system lists the affected appointments and asks me to confirm; after confirmation the appointments are cancelled and the patients are notified. AC4 (error case) Given the end of the period is before its start, when I save it, then the system rejects the input and shows a validation message.		
Notes / open questions	Who reschedules the patients affected by a new unavailability – the doctor, the receptionist or the system automatically?		

US-04 – Register a new doctor

ID	US-04	**Author**	Nazar Martyniuk
Title	Register a new doctor	**Priority**	Must
Related requirements	FR-03, FR-20	**Estimate (optional)**	3 story points
Story	As a clinic administrator, I want to register a new doctor with their specialty and working hours, so that patients can start booking appointments with them.		
Acceptance criteria	AC1 Given I am logged in as the clinic administrator, when I enter the doctor's name, specialty, phone, e-mail and weekly working hours and save, then the doctor appears in the search results for that specialty and their working hours appear as free slots. AC2 Given I have registered a doctor, when I save, then a user account with the role Doctor is created for them and login details are sent to their e-mail address. AC3 (error case) Given a doctor with the same e-mail address already exists, when I save, then the system refuses the registration and shows which field is duplicated. AC4 (error case) Given I leave the specialty or the working hours empty, when I save, then the system rejects the form and marks the missing fields.		
Notes / open questions	Can a doctor have more than one specialty?		

6. Open issues

Questions still waiting for the client's answer. Close them in the next meetings and record the decision.

No.	Open issue	Raised by	Status / decision
1			

Part II – Analysis, design and testing

Leave Part II empty in Assignment 2. Each section is completed in the assignment named in its heading – the assignment description tells you exactly what and how much to add. Earlier sections may be updated later; record every change in the revision history. Use the IDs from Part I (FR-, NFR-, US-) to show where each element comes from.